



Internet Basics

Browsers, search engines, security, and email – the most important digital skills explained concisely.

DIGITAL LITERACY

Browser vs. Search Engine – The Difference

Many people confuse these two terms. Yet, the distinction is fundamental:

Web Browser

A **program** for opening websites – your "window to the web".

Examples: Chrome, Firefox, Edge, Safari

Search Engine

An **internet service** for searching information – runs inside the browser.

Examples: Google, Bing, DuckDuckGo

The order is always: first open the browser, then use a search engine within it.

Opening and using a browser

On your computer, click an icon in the taskbar, on the desktop, or in the start menu. On your smartphone, tap the browser app.

At the top of the browser, you will see the **address bar**. There you can:

Enter a web address

e.g., www.wikipedia.org

Type a search term

e.g., Berlin weather

Modern browsers allow both in the same bar.



Key Parts of a Browser



Address Bar

Navigate to a website or enter a search term



Back / Forward

Navigate between visited pages



Reload

Refresh the page if content is not displaying correctly

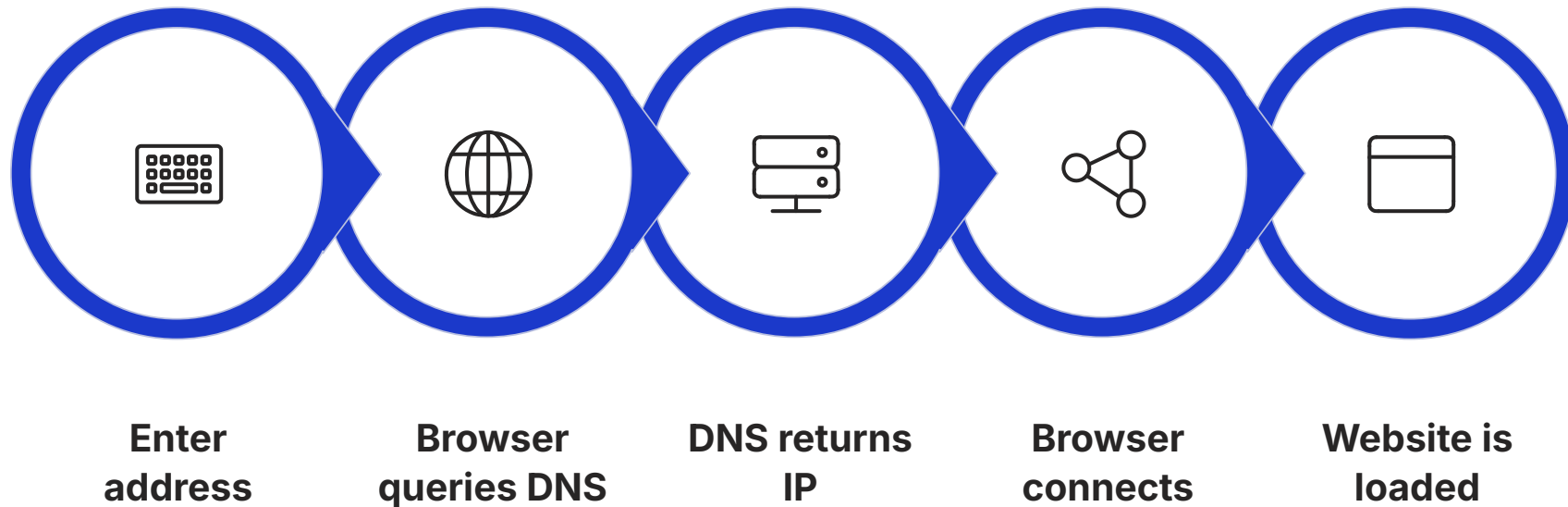


Tabs & Bookmarks

Keep multiple pages open simultaneously and save your favorite sites

What happens when you open a website?

There is more going on in the background than you might think. The **Domain Name System (DNS)** translates a name like *example.com* into a technical IP address so that the browser can find the correct server.



Only once the server has been found does the browser request the content and display it as a readable website.

Tabs, Bookmarks, History and Downloads

Tabs

Keep multiple pages open at the same time – practical, but: the more tabs, the faster you lose track. Less is often more.

Bookmarks

Save important pages (learning portals, authorities, online banking) for quick access.

History

Shows visited pages – helpful if you want to find a page you accidentally closed.

Downloads

Shows downloaded files (PDFs, images, forms). Be especially careful here: **not every downloaded file is safe.**

Using Search Engines Effectively

A search engine does not search the entire internet in real time; instead, it works with a pre-built **index**—a massive collection of captured webpages.

You enter search terms and receive a list of results that best match your query.

- ❏ Good research begins with a good search query. The more precise your search terms, the better the results.



Formulating Effective Search Queries

Many people search too broadly and get many results, but not the right ones.

→ Concrete instead of vague

Instead of *laptop*, better: **laptop for college under 800 dollars**

→ Quotation marks for exact phrases

"digital literacy" – searches for this exact phrase

→ Minus sign to exclude

jaguar -car – excludes the word "car" from the results

→ site: for a specific website

privacy site:bsi.bund.de – searches only on this domain

How to read and evaluate search results

A search result consists of a **title**, **URL**, and **description**. A page at the very top is not automatically the best or most trustworthy one.

Who is the publisher?

Official authority, university, company, or private blog?

Is the information current?

Outdated content can be misleading.

Does the page match the query?

Particularly reliable: government websites, universities, reputable professional portals.

From Finding to Understanding

For effective learning, it is not enough to quickly Google an answer. If you just copy a term, you will quickly forget it. If you can explain it in your own words, you have truly understood it.

Gain an overview

Clarify terms

Compare multiple sources

Explain in your own words

CHAPTER 2

Staying safe online

The internet is useful, but not every site and not every message is trustworthy. Safety doesn't mean being distrustful of everything – but rather **being attentive, recognizing warning signs, and developing good habits.**



Recognizing Safe and Unsafe Websites

✓ Trustworthy Features

- **https://** and padlock icon in the address bar
- Clean, understandable URL
- Clear provider information
- No aggressive popups
- Official domain or well-known organization

⚠ Warning Signs

- Strange address with typos (e.g., *paypal.com*)
- Urgency: "Only 3 minutes left!"
- Extreme promises: "Free iPhone"
- Browser warnings or certificate issues
- Automatically starting downloads

📌 **Important:** HTTPS alone does not mean a site is legitimate – fraudulent sites can also use HTTPS.

The URL – where forgeries hide

Don't just look at a site's logo or design; **always check the address**. That is exactly where many forgeries hide.

✓ **Genuine**

<https://www.sparkasse.de>

⚠ **Suspicious**

<https://sparkasse-login-sicher.net>

Phishing attacks often use fake login pages and messages that create pressure: *"Your account will be locked if you don't click immediately."* You should **never just dismiss** browser warnings – they are there to protect you.

Data Privacy in Everyday Life

Do not share more data than necessary – that is the most important ground rule. Many websites and apps collect information about your behavior and interests.

Share personal data sparingly

Address, phone number, date of birth, ID documents, bank details, and passwords only when absolutely necessary.

Handle cookies mindfully

Do not automatically accept everything.
Carefully read consent windows.

Private mode ≠ anonymous

Incognito mode only prevents local storage. Websites, employers, or internet service providers can still see your behavior.

Safe Surfing Habits

01

Access important sites directly

Use known addresses or bookmarks – not links from messages.

02

Download programs only from official sites

No downloads from unknown sources.

03

Log out on shared devices

Never save passwords on public devices.

04

Perform updates regularly

Security updates close vulnerabilities. An outdated browser is easier to attack.

05

Be careful with public Wi-Fi

Prefer to do sensitive things like online banking on trusted networks.

The most important protection rule

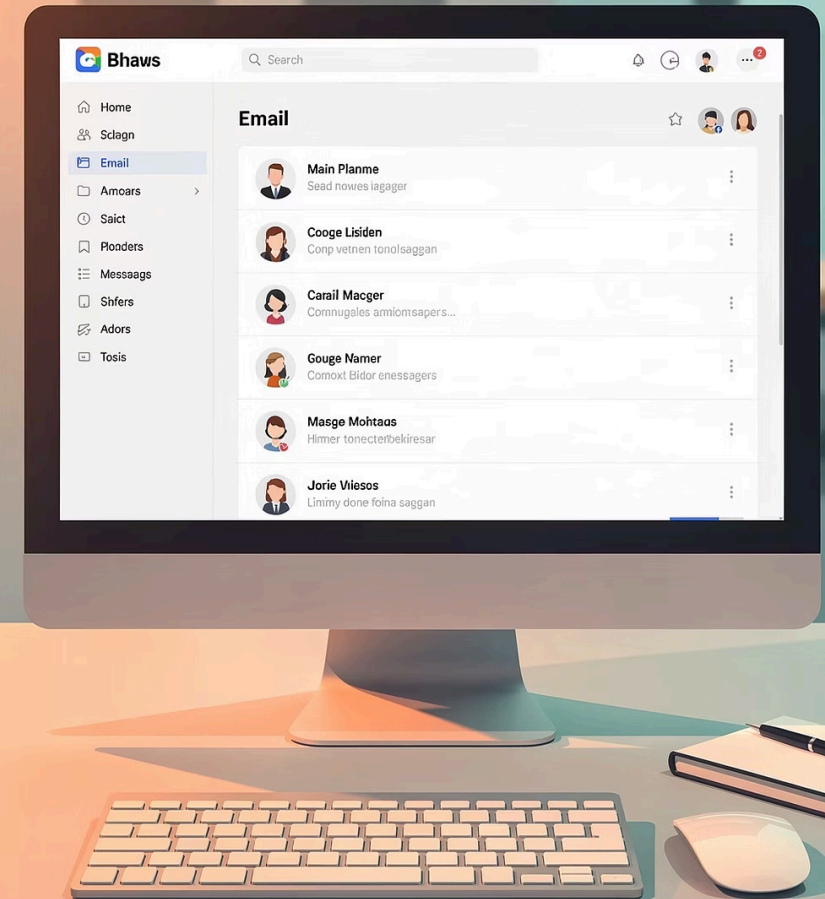
If something on the internet seems **too urgent, too good, or too strange**, taking a pause is the best security measure. Fraud often works because people act quickly under pressure. Anyone who stops briefly and checks protects themselves significantly better.



CHAPTER 3

Email Basics

Emails are among the most important basic digital skills—for private communication, school, university, work, contact with authorities, and job applications. An email functions like a digital letter: faster, but structured similarly.



Email Structure



From / To / Cc / Bcc

Sender, primary recipient, visible and invisible secondary recipients



Message Body

The actual content with greeting, introduction, main body, and closing



Subject

Short, specific headline – e.g., *Appointment confirmation for Monday*. Not: *Hello!!!*



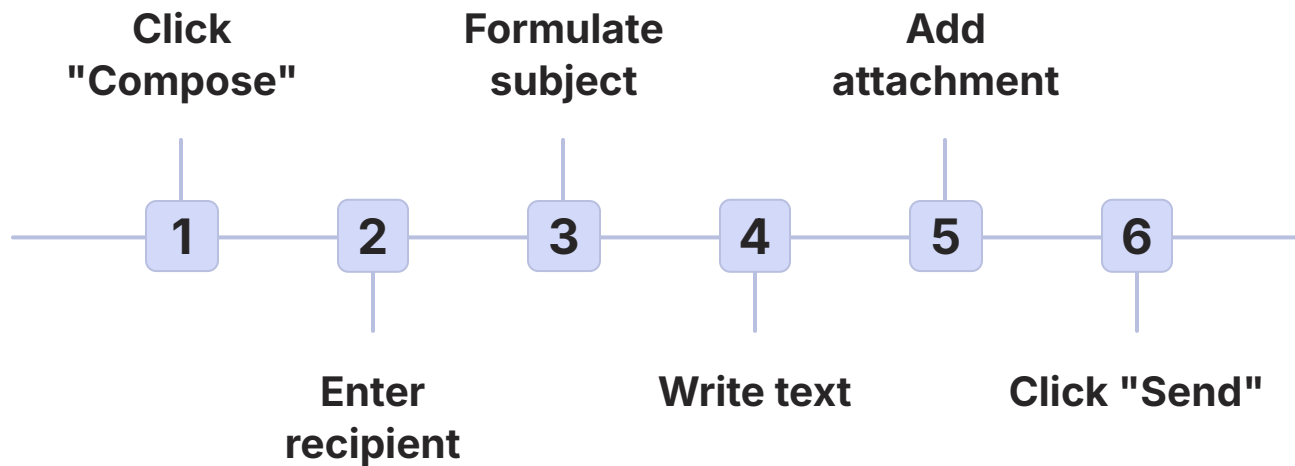
Attachments & Signature

Additional files as well as name and contact details at the end

PROCESS

Writing and sending an email

Writing follows the same simple process every time – once you understand it, you can use almost any mail service.



❏ Only when you click **"Send"** is the message actually transmitted.

Receiving, replying to, and forwarding emails

Incoming messages arrive in your **Inbox**. Important: It is also worth checking your **Spam folder** occasionally – legitimate emails can sometimes end up there.

Reply

Write back only to the sender

Reply all

Reply to all original recipients – **use with caution!**

Forward

Send the email to another person

You should use **"Reply all"** deliberately – otherwise, you might send your message to more people than necessary.

Caution with suspicious emails

Not every email is harmless. Typical signs of **phishing messages**:

Unknown sender

You do not know the person or organization.

Pressure and urgency

"Your account will be blocked!" – immediate action is demanded.

Suspicious links or attachments

Bank details, passwords, or codes are requested.

- ❏ **Protection rule:** Do not click on sensitive links directly from the email – instead, access important services directly in your browser.



Adding and saving attachments

Add attachment

Click the **paperclip icon**, select a file, and add it. Before sending, check whether you have attached the right file.

Save attachment

Save received files consciously in folders, e.g.
Documents/Applications – this saves a lot of search time later.

Important file types

- **.pdf** – Forms, documents, articles
- **.jpg / .png** – Photos, screenshots
- **.docx** – Letters, reports
- **.xlsx** – Lists, tables
- **.pptx** – Presentations
- **.zip** – Multiple files packed

Don't open attachments blindly

Attachments are a classic path for malware and fraud attempts. Do not open attachments if:

- ☐ You don't know the sender
- ☐ You weren't expecting the message
- ☐ The file name looks strange
- ☐ You are being pressured to act quickly

Be especially careful with executable files or unknown archive files. When in doubt: **check first, click later.**

Why digital basic literacy is so important

Anyone who can confidently navigate browsers, search engines, website verification, and email already masters a large part of practical digital literacy.

Find information reliably

Better identify risks

Communicate clearly

Use digital tools consciously

These skills are not just "computer knowledge" – they are a foundation for self-determined action in the digital world.

